



What's prompt engineering?

Prompt Engineering

1. Definition

Prompt engineering is the disciplined practice of designing inputs that steer Large Language Models (LLMs) toward **specific, high-quality outputs**. A good prompt acts as a blueprint, translating human intent into a form the model can understand and obey.

2. Why It Matters

- **Control:** Precise prompts reduce hallucinations and off-topic drift.
 - **Efficiency:** Well-structured instructions cut down on trial-and-error, saving time and tokens.
 - **Customization:** Prompts let you tailor tone, format, and depth without retraining the model.
 - **Accessibility:** Non-developers can leverage powerful AI capabilities through natural language alone.
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3. Core Principles

Principle	Practical Meaning
Clarity	Use unambiguous language; avoid vague verbs like "talk about."
Specificity	Spell out format, length, audience, and stylistic constraints.
Context	Supply only the information the model truly needs—no noise.
Structure	Organize instructions (e.g., headings, bullet lists) so the model can parse intent easily.
Iteration	Test → observe output → refine prompt; repeat until goals are met.

4. Typical Prompt-Engineering Workflow

1. **Define the goal** – What does “success” look like?
 2. **Draft an initial prompt** – Include role, task, context, and constraints.
 3. **Generate and evaluate** – Run the prompt; assess relevance, accuracy, tone.
 4. **Refine** – Adjust wording, add examples, tweak decoding settings (temperature, top-p, etc.).
 5. **Automate or template** – Once stable, turn the prompt into a reusable component or API call.
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5. Key Techniques

5.1 Role & Persona Framing

“You are a senior UX researcher...”

Sets voice and domain expertise.

5.2 Zero-Shot, Few-Shot, & Multi-Shot

- **Zero-Shot:** No examples; rely on clear instructions.
- **Few-Shot:** Provide 2-3 exemplars to enforce style or reasoning pattern.
- **Multi-Shot / Chain-of-Thought:** Offer longer step-by-step examples for complex tasks.

5.3 Constraint Setting

- Word or character limits
- Output structure (JSON, table, Markdown)
- Tone or reading level

5.4 Self-Evaluation & Reflection

Ask the model to critique or improve its own answer before finalizing.

5.5 External Validation

Pair prompts with regex checks, knowledge-base queries, or human review loops.

6. Skills of a Prompt Engineer

- **Linguistic precision** – Craft concise, domain-appropriate instructions.
 - **Analytical mindset** – Diagnose failure modes and iterate systematically.
 - **LLM mechanics** – Know how temperature, top-k, and top-p affect decoding.
 - **Domain knowledge** – Provide relevant context and detect subtle inaccuracies.
 - **UX sensibility** – Balance cognitive load for both the model and end user.
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7. Common Pitfalls

1. **Under-specification:** Vague prompts produce vague answers.
 2. **Over-specification:** Excessive rules can overwhelm or conflict.
 3. **Context bloat:** Large, irrelevant context dilutes prompt focus and wastes tokens.
 4. **Ignoring decoding settings:** Great wording + wrong temperature can still fail.
 5. **One-off mindset:** Failing to treat prompts as assets that deserve version control and documentation.
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8. Best-Practice Checklist

- ☐ State the *role/persona* the model should adopt.
 - ☐ Describe the *task* in a single, direct sentence.
 - ☐ Provide *essential context*—nothing more.
 - ☐ Set *format, tone, and length* constraints explicitly.
 - ☐ Include *few-shot examples* if style consistency is critical.
 - ☐ Test with varying *decoding parameters* to find the sweet spot.
 - ☐ Iterate based on measurable success criteria.
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9. Key Takeaways

- Prompt engineering turns raw model potential into **practical, reliable solutions**.
- Success hinges on clarity, specificity, structure, and iterative refinement.
- Mastering prompt techniques—and the decoding knobs behind them—lets you channel LLM power toward virtually any text-based task with confidence.